

Learning and Sparse Control of Multi-agent Systems

Dohyun Kim

Seoul National University

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Introduction

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Overview

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 - Theorem 3.10 (Sparse feedback control)

Multi-agent systems

Fundamental Goal : To reveal the possible relationship between the simple binary forces acting at individual level and the potential emergence of a global behavior in the form of specific patterns.

Equation :

$$\left\{ \begin{array}{l} \dot{x}_i = v_i \\ \dot{v}_i = (S + H * \mu_N)(x_i, v_i) \\ i = 1, \dots, N, t \in [0, T] \\ \text{where, } \mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{(x_j, v_j)} \end{array} \right.$$

The meaning of the equation depends on the choice of S and H .

Alignment models as proposed by Cucker and Smale

In this case,

$$S \equiv 0 \quad \text{and} \quad H(x, v) = a(\|x\|)(-v)$$

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N a(\|x_i - x_j\|)(v_j - v_i), \quad i = 1, \dots, N. \\ \text{where } a(r) = \frac{1}{(1 + r^2)^\beta} \end{cases}$$

or in matrix form, we can rewrite as

$$\begin{cases} \dot{x} = v \\ \dot{v} = -L^a(x)v \end{cases}$$

where the matrix $L^a(x)$ is the Laplacian of the $N \times N$ adjacency matrix $(a(\|x_i - x_j\|)/N)_{i,j=1}^N$

Some facts of C-S model (Consensus)

Define the symmetric bilinear form

$$B(u, v) = \frac{1}{2N^2} \sum_{i,j} \langle u_i - u_j, v_i - v_j \rangle = \frac{1}{N} \sum_{i=1}^N \langle u_i, v_i \rangle - \langle \bar{u}, \bar{v} \rangle,$$

and the quantities

$$X(t) = B(x(t), x(t)), \quad V(t) = B(v(t), v(t)),$$

Theorem 3.1

Let $(x_0, v_0) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ be such that $X_0 = B(x_0, x_0)$ and $V_0 = B(v_0, v_0)$ satisfy

$$\gamma(X_0) := \sqrt{N} \int_{\sqrt{NX_0}}^{\infty} a(\sqrt{2}r) dr > \sqrt{V_0}.$$

Then the solution of the C-S system with initial data (x_0, v_0) tends to the consensus.

Some facts of C-S model (Example of Nonconsensus)

Example 3.2

Consider $N = 2, d = 1, \beta = 1$ and

$x(t) = x_1(t) - x_2(t), v(t) = v_1(t) - v_2(t)$ which is relative position and velocity of two agents on the real line. Then (x, v) satisfies

$$\begin{cases} \dot{x} = v \\ \dot{v} = -\frac{v}{1+x^2} \end{cases}$$

with initial conditions $x(0) = x_0$ and $v(0) = v_0 > 0$. By direct integration

$$v(t) = -\arctan x(t) + \arctan x_0 + v_0$$

Hence, if $\arctan x_0 + v_0 + \frac{\pi}{2} > \epsilon$, then we have

$$v(t) > \frac{\pi}{2} + \epsilon - \arctan x(t) > \epsilon, \quad \forall t \in \mathbb{R}_+$$

Consensus region

Goal : To find **admissible controls** steering the system to **the consensus region** starting from any possible initial condition (in finite time).

Definition (Consensus region)

We call the set of $(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ satisfying

$$\int_{\sqrt{X_0}}^{\infty} a(\sqrt{2Nr})dr \geq \sqrt{V_0} \quad (1)$$

the consensus region.

Note that if the integral $\int_0^{\infty} a(r)dr$ diverges, then every pair (X, V) is in consensus region. Hence we will assume $a \in L^1(0, \infty)$.

Cucker-Smale model with control

From now on, we will consider the following model:

$$\left\{ \begin{array}{l} \dot{x}_i(t) = v_i(t) \\ \dot{v}_i(t) = \frac{1}{N} \sum_{j=1}^N a(\|x_j(t) - x_i(t)\|)(v_j(t) - v_i(t)) + u_i(t) \\ \text{with the constraint } \sum_{i=1}^N \|u_i(t)\| \leq M \end{array} \right.$$

Here the norm $\|u_i\|$ is $L^2(\mathbb{R}^d)$ -norm and the constant $M > 0$ represents the budget of strength of the control force which we can apply on the system and likely, M is supposed to be very small.

Existence of total control

As a first observation, we prove that an appropriate choice of the control law can always stabilize the system to consensus.

Proposition 3.3 (Total Control)

For any initial data $(x_0, v_0) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ and $M > 0$, there exists $T > 0$ and $u : [0, T] \rightarrow (\mathbb{R}^d)^N$, with $\sum_{i=1}^N \|u_i(t)\| \leq M$ for every $t \in [0, T]$ such that the associated solution reaches the consensus region in time $T > 0$.

Note that this simple result is somehow remarkable in this way : Not only it shows that we can steer the system to consensus from **any** initial condition, but that the strength of the control $M > 0$ can be arbitrarily small.

Proof of Proposition 3.3

(proof)

$$\begin{aligned}
 \frac{d}{dt} V(t) &= \frac{d}{dt} B(v(t), v(t)) \\
 &= -2B(L^a(x)v, v) + 2B(u, v) \\
 &\leq 2B(u(t), v(t)) = -2\alpha B(v_\perp, v) = -2\alpha V(t)
 \end{aligned}$$

Note that $(\mathbb{R}^d)^N$ can be decomposed as two orthogonal spaces:

$$(\mathbb{R}^d)^N = \mathcal{V}_f \oplus \mathcal{V}_\perp$$

where

$$\mathcal{V}_f := \{(v_1, \dots, v_N) \in (\mathbb{R}^d)^N : v_1 = \dots = v_N\}$$

$$\mathcal{V}_\perp := \{(v_1, \dots, v_N) \in (\mathbb{R}^d)^N : \sum_{i=1}^N v_i = 0\}$$

Hence for $v \in (\mathbb{R}^d)^N$, we can write $v = v_f + v_\perp$ with $v_f \in \mathcal{V}_f, v_\perp \in \mathcal{V}_\perp$.

Drawback of total control

However, this result has perhaps only theoretical validity, because the chosen stabilizing control $u = -\alpha(v - \bar{v}) = -\alpha v_{\perp}$ needs to act instantaneously on all the agents and needs to be informed of the status of the entire system. (This is why we call this u 'total control'.) In other words, this control is not convenient for practical purposes since it requires to act at every instant of time on all the agents in order to steer the system to consensus, which may require a large amount of instantaneous communications.

- ⇒ We need more economical and practical(sparse) controls!
- ⇒ Good strategy to steer the system to consensus is actually the minimization of $B(u(t), v(t))$ w.r.t u , for all t !

Existence of greedy control

Theorem 3.4

For every initial condition $(x_0, v_0) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ and $M > 0$, there exist $T > 0$ and a sparse piecewise constant control $u : [0, T] \rightarrow (\mathbb{R}^d)$, with $\sum_{i=1}^N \|u_i\| \leq M$ for every $t \in [0, T]$ such that associated (AC) solution reaches the consensus region at the time T . More precisely, we can choose adaptively the control law explicitly as one of the solutions of the 'variational problem'.

This choice of the control makes $V(t) = B(v(t), v(t))$ vanishing in finite time, in particular there exists $T > 0$ such that

$$B(v(t), v(t)) \leq \gamma(x)^2, t \geq T.$$

Note that this result holds independently of the initial conditions and of the strength of the control M . Moreover, it will be turned out that this theorem indicates the controllability of consensus systems simply by acting on one agent at each time and for nontrivial intervals of time.

Variational problem (Componentwise sparsity)

Definition

For every $M > 0$ and every $(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$, let $U(x, v)$ be defined as the set of solutions of the variational problem

$$\min \left(B(v, u) + \gamma(B(x, x)) \frac{1}{N} \sum_{i=1}^N \|u_i\| \right) \quad \text{subject to} \quad \sum_{i=1}^N \|u_i\| \leq M$$

$$\text{where } \gamma(X) = \int_{\sqrt{X}}^{\infty} a(\sqrt{2Nr}) dr$$

- minimizing $B(v, u) = B(v_{\perp}, u)$ means that, at every instant of time, the control $u \in U(x, v)$ is of the form $u = -\alpha \cdot v_{\perp}$, for some $\alpha = (\alpha_1, \dots, \alpha_N)$
- $\sum_{i=1}^N \|u_i\|$ has the function of enforcing **sparsity**, i.e., most of the α_i 's will turn out to be zero.
- the threshold $\gamma(X)$ is chosen in such a way that when the control switches-off, the criterion (1) is fulfilled.

Componentwise sparsity

Note that for every $(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ and every element $u(x, v) \in U(x, v)$, there exist non-negative real numbers α_i 's s.t.

$$u_i(x, v) = \begin{cases} 0 & \text{if } v_{\perp i} = 0, \\ -\alpha_i \frac{v_{\perp i}}{\|v_{\perp i}\|} & \text{if } v_{\perp i} \neq 0, \end{cases}$$

where $0 \leq \sum_{i=1}^N \alpha_i \leq M$.

Decomposition of $(\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$

$(\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ can be partitioned in the union of the four disjoint subsets $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ and $\mathcal{C}_4$ defined as follows:

$$\mathcal{C}_1 := \{(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N \mid \max_{1 \leq i \leq N} \|v_{\perp i}\| < \gamma(B(x, x))\}$$

$$\mathcal{C}_2 := \{(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N \mid \max_{1 \leq i \leq N} \|v_{\perp i}\| = \gamma(B(x, x))\}$$

$$\mathcal{C}_3 := \{(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N \mid \max_{1 \leq i \leq N} \|v_{\perp i}\| > \gamma(B(x, x))$$

and there exists a unique $i \in \{1, \dots, N\}$ such that $\|v_{\perp i}\| > \|v_{\perp j}\|$ for every $j \neq i\}$

$$\mathcal{C}_4 := \{(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N \mid \max_{1 \leq i \leq N} \|v_{\perp i}\| > \gamma(B(x, x))$$

and there exists $k \geq 2$ and $i_1, \dots, i_k \in \{1, \dots, N\}$ such that $\|v_{\perp i_1}\| = \dots = \|v_{\perp i_k}\|$ and $\|v_{\perp i_1}\| > \|v_{\perp j}\|$ for $j \neq \{i_1, \dots, i_k\}$

Control u related to $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ and $\mathcal{C}_4$

- For every $(x, v) \in \mathcal{C}_1 \cup \mathcal{C}_3$, the set $U(x, v)$ is single-valued and its value is a sparse vector with at most one nonzero component. More precisely,

$$U|_{\mathcal{C}_1} = \{0\} \quad \text{and} \quad U|_{\mathcal{C}_3} = \{(0, \dots, 0, -Mv_{\perp_i}/\|v_{\perp_i}\|, 0, \dots, 0)\}$$

and note that $(\mathcal{C}_1 \cup \mathcal{C}_3)^c$ has Lebesgue measure zero in $(\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$.

- For every $(x, v) \in \mathcal{C}_2 \cup \mathcal{C}_4$, then a control in $U(x, v)$ may not be sparse: indeed, the set $U(x, v)$ consists of all $u = (u_1, \dots, u_N) \in (\mathbb{R}^d)^N$ such that $u_i = -\alpha_i v_{\perp_i}/\|v_{\perp_i}\|$ for every $i = 1, \dots, N$ s.t. $0 \leq \sum_{i=1}^N \alpha_i \leq M$ if $(x, v) \in \mathcal{C}_2$ and $\sum_{i=1}^N \alpha_i = M$ if $(x, v) \in \mathcal{C}_4$.

Proof of Theorem 3.4

(proof) For the proof, we first need the positively invariant property of the $\mathcal{C}_1$ along the Cucker-Smale dynamics [Appendix Lemma A.4., 13].

Lemma A.4., [13]

Let $(x(t), v(t))$ be a solution of Cucker-Smale system. Then for every $t \geq 0$, we have

$$\frac{d}{dt} \left(\max_{1 \leq i \leq N} \|v_{\perp_i}\| - \gamma(B(x(t), x(t))) \right) \leq 0$$

Sparse is better - Optimality

Since this control strategy is designed to act on at most one agent at each time, we claim that in some sense it is instantaneously the 'best one'. To clarify this notion of 'instantaneous optimality' which also explains its 'greedy' nature, we will compare this strategy with all other control strategies $u(x, v) \in U(x, v)$ and discuss their efficiency in terms of the decay rate of the functional V .

Optimality of greedy control

Theorem 3.5

Consider 'generic' control u of components

$$u_i(x, v) = \begin{cases} 0 & \text{if } v_{\perp_i} = 0 \\ -\alpha_i \frac{v_{\perp_i}}{\|v_{\perp_i}\|} & \text{if } v_{\perp_i} \neq 0 \end{cases}$$

where $\alpha_i \geq 0$ such that $\sum_{i=1}^N \alpha_i \leq M$. The 1-sparse control is the minimizer of $dV(t)/dt$ among all the control of the previous form. In other words, this control strategy is the best choice in terms of the rate of convergence to consensus.

Proof of Theorem 3.5

(proof) We select the componentwise sparse feedback control $u = u(x, v) \in U(x, v)$ according to the following criterion:

- if $\max_{1 \leq i \leq N} \|v_{\perp_i}\| \leq \gamma(B(x, x))^2$, then $u = 0$.
- if $\max_{1 \leq i \leq N} \|v_{\perp_i}\| > \gamma(B(x, x))^2$, let $j \in \{1, \dots, N\}$ be the smallest index

such that $\|v_{\perp_j}\| = \max_{1 \leq i \leq N} \|v_{\perp_i}\|$, then $u_j = -M \frac{v_{\perp_j}}{\|v_{\perp_j}\|}$

and $u_i = 0$, for every $i \neq j$.

Repulsion-attraction models proposed by Cucker and Dong

In this case,

$$S(x, v) = -bv \quad \text{and} \quad H(x, v) = a(\|x\|^2)(-x) - f(\|x\|^2)(-x)$$

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = -b_i v_i + \sum_{j=1}^N (a - f)(\|x_i - x_j\|^2)(x_j - x_i), \quad i = 1, \dots, N. \\ \text{where } a(r) = \frac{1}{(1 + r^2)^\beta}, f(r) = \frac{1}{r^\delta} \text{ and } t \mapsto b_i(t) \in [0, \Lambda] \end{cases}$$

Here a describes attraction and b describes repulsion.

Some facts on C-D model (cohesion)

Define the (total energy) functional $E(t) := E(x, v)$ and the energy threshold θ

$$E(t) := \sum_{i=1}^N \|v_i(t)\|^2 + \frac{1}{2} \sum_{i < j}^N \left(\int_0^{\|x_i - x_j\|^2} a(r) dr + \int_{\|x_i - x_j\|^2}^{+\infty} f(r) dr \right),$$

$$\theta := \frac{N-1}{2} \int_0^{+\infty} a(r) dr$$

Theorem 3.7 [30]

If $E(0) < \theta$, then there exist finite constants $c_0 > 0$ and $C_0 > 0$ such that

$$c_0 \leq \|x_i(t) - x_j(t)\| \leq C_0, \quad \text{for all } t \geq 0.$$

In other words, the systems converge autonomously to the stable configuration of keeping confined and collision avoiding in space, uniformly in time

Some facts on C-D model (Non-cohesion example)

Example 3.8

Consider $N = 2$, $d = 1$, $\beta > 1$, $f \equiv 0$, $b_i \equiv 0$ and $x(t) = x_1(t) - x_2(t)$, $v(t) = v_1(t) - v_2(t)$ which are relative position and velocity of two agents on the real line. Then (x, v) satisfies

$$\begin{cases} \dot{x} = v \\ \dot{v} = -\frac{x}{(1+x^2)^\beta}. \end{cases}$$

If the initial conditions satisfy $x(0) > 0$ and $v(0) > 0$ and

$$v(0)^2 \geq \frac{1}{(\beta-1)(1+x(0)^2)^{\beta-1}},$$

then $x(t) \rightarrow \infty$ as $t \rightarrow \infty$.

Some facts on C-D model (Non-cohesion example)

Example 3.8 (Continued)

Moreover, by direct calculation, for all t ,

$$v(t)^2 = \frac{1}{(\beta - 1)(1 + x(t)^2)^{\beta-1}} + \left[v(0)^2 - \frac{1}{(\beta - 1)(1 + x(0)^2)^{\beta-1}} \right] > 0$$

and $x(t) \rightarrow \infty$ as $t \rightarrow \infty$.

Cucker-Dong model with control

Goal : Recall that if $E(0) < \theta$, then the system keeps cohesive and collision avoiding. Hence, one can pose the question of whether, given $E(0) > \theta$, a **sparse** control can bring in finite time T the energy under the threshold $E(T) < \theta$.

Definition of Sparse control of Cucker-Dong model

Definition 3.9

Let $(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ and $0 \leq \epsilon \leq \frac{M}{E(0)}$. Define the **sparse feedback control** $u(x, v) = [u_1(x, v), \dots, u_N(x, v)]^T \in (\mathbb{R}^d)^N$ associated to (x, v) as

$$u_i(x, v) = \begin{cases} -\epsilon E(x, v) \frac{v_i}{\|v_i\|} & \text{if } i = \hat{l}(x, v), \\ 0 & \text{if } i \neq \hat{l}(x, v), \end{cases}$$

where $\hat{l}(x, v)$ is the minimal index such that

$$\|v_{\hat{l}(x, v)}\| = \max_{j=1, \dots, N} \|v_j\|$$

Existence of sparse feedback control

Theorem 3.10

There exist a constant $\eta > 1$ and a finite time $T > 0$ such that if

$$\theta < E(0) < \eta\theta$$

then the energy of controlled system with the above sparse feedback control of Definition 3.9 assumes

$$E(T) < \theta$$

Hence, for $t \geq T$, the uncontrolled system keeps cohesive and collision avoiding.

Note that the initial conditions need to fulfill a second threshold to be sparsely controlled by a sparse feedback control in Definition 3.9.

Proof of Theorem 3.10

More precisely, we want to prove the following theorem:

Theorem 3.10' [7]

Suppose that $\|\bar{v}(0)\| \geq \eta > 0$, for η small enough, and that $E(0) > \theta$. Then if following conditions hold:

$$\log\left(\frac{E(0)}{\theta}\right) \leq \frac{N(\|\bar{v}(0)\|^2 - \eta^2)}{2E(0)},$$

$$\frac{2N\Lambda E(0) \log(E(0)/\theta)}{N(\|\bar{v}(0)\|^2 - \eta^2) - 2E(0) \log(E(0)/\theta)} \leq \epsilon \leq \frac{2M\eta}{E(0)},$$

then the control designed in Definition 3.9' satisfies $\sum_{i=1}^N \|u_i(t)\| \leq M$ and forces $E(t) \leq \theta$ as soon as $t > T^*$ where

$$T^* := \frac{1}{\epsilon} \log\left(\frac{E(0)}{\theta}\right)$$

Sketch of proof of Theorem 3.10'

(Sketch of the proof) First, define the auxiliary time $T = \frac{N(\|\bar{v}(0)\|^2 - \eta^2)}{(2N\lambda + \epsilon)E(0)}$.

(• Step 1) $E(t)$ is (exponentially) decreasing in $[0, T]$. Hence T^* is the time when $E(t)$ touches θ .

(• Step 2) Also, the control u keeps satisfying the constraint $\sum_{i=1}^N \|u_i(t)\| \leq M$ for all $t \in [0, T]$.

(• Step 3) $T^* \leq T$.

Proof of Step 1..

Claim $\frac{d}{dt}E(x(t), v(t)) \leq 2 \sum_{i=1}^N \langle u_i(t), v_i(t) \rangle$ for all $t \geq 0$. Actually we need definition of (scaled) sparse feedback control as below:

Definition 3.9'

Let $(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ and $0 \leq \epsilon \leq \frac{M}{E(0)}$. Define the **sparse feedback control** $u(x, v) = [u_1(x, v), \dots, u_N(x, v)]^T \in (\mathbb{R}^d)^N$ associated to (x, v) as

$$u_i(x, v) = \begin{cases} -\epsilon E(x, v) \frac{v_i}{2\|v_i\|^2} & \text{if } i = \hat{i}(x, v), \\ 0 & \text{if } i \neq \hat{i}(x, v), \end{cases}$$

where $\hat{i}(x, v)$ is the minimal index such that

$$\|v_{\hat{i}(x, v)}\| = \max_{j=1, \dots, N} \|v_j\|$$

Proof of Step 2..

Claim If $\|\bar{v}(0)\| \geq \eta > 0$, then we have $\|v_{\iota(x,v)}(t)\| \geq \eta$ for all $t \in [0, T]$.
(proof of the claim) [Lemma 7, [7]].

Then if

$$\epsilon \leq \frac{2M\eta}{E(0)},$$

then we have

$$\sum_{i=1}^N \|u_i(t)\| \leq M \quad \text{for all } t \in [0, T].$$

Proof of Step 3..

Claim $T^* \leq T$.

Note that

$$\begin{aligned}
 T^* \leq T &\iff \frac{1}{\epsilon} \log \left(\frac{E(0)}{\theta} \right) \leq \frac{N(\|\bar{v}(0)\|^2 - \eta^2)}{(2N\Lambda + \epsilon)E(0)} \\
 &\iff \epsilon \left[N(\|\bar{v}(0)\|^2 - \eta^2) - 2E(0) \log \left(\frac{E(0)}{\theta} \right) \right] \geq 2N\Lambda E(0) \log \left(\frac{E(0)}{\theta} \right) \\
 &\iff \epsilon \geq \frac{2N\Lambda E(0) \log \left(\frac{E(0)}{\theta} \right)}{N(\|\bar{v}(0)\|^2 - \eta^2) - 2E(0) \log \left(\frac{E(0)}{\theta} \right)}
 \end{aligned}$$

The last equivalence follows from the assumption of the theorem which says that the quantity between square brackets is positive. Finally since we assume the last equivalence, we have $T^* \leq T$.

The End

Thank you !